



KILO & MILLI

Understanding Units in Electronics

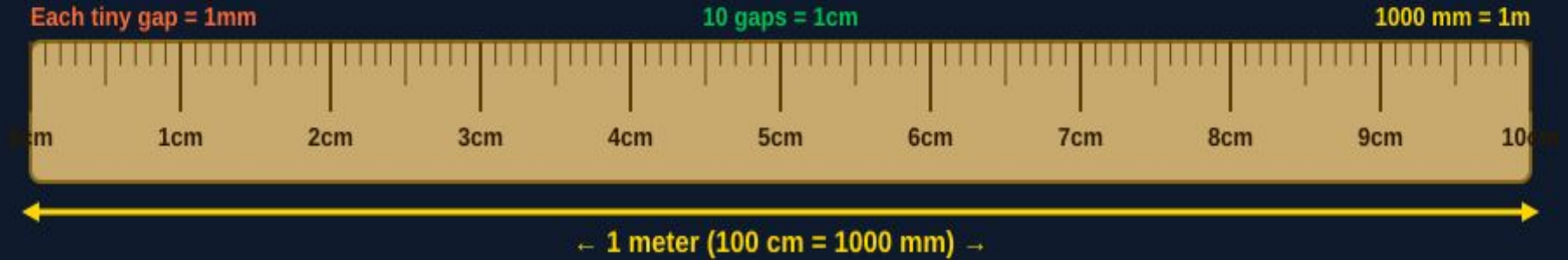
millivolts · milliamps · kilohms · and more

You already know this!

You use milli and kilo every day.
millimeters → meters → kilometers
Same idea. Different units.

Start Here: The Meter Stick

How many millimeters in 1 meter? Let's count.



1000 millimeters = 1 meter · 1000 meters = 1 kilometer · milli always means $\div 1000$

10 mm = 1 cm

100 cm = 1 m

1000 mm = 1 m

1000 m = 1 km

The pattern: milli means $\div 1000$. So 1000 millimeters = 1 meter. 1000 milliamps = 1 amp. Same rule every time.

The SI Prefix Ladder — Vocabulary

Every unit of measurement uses the same prefixes. Most adults only know 2 or 3 of these.

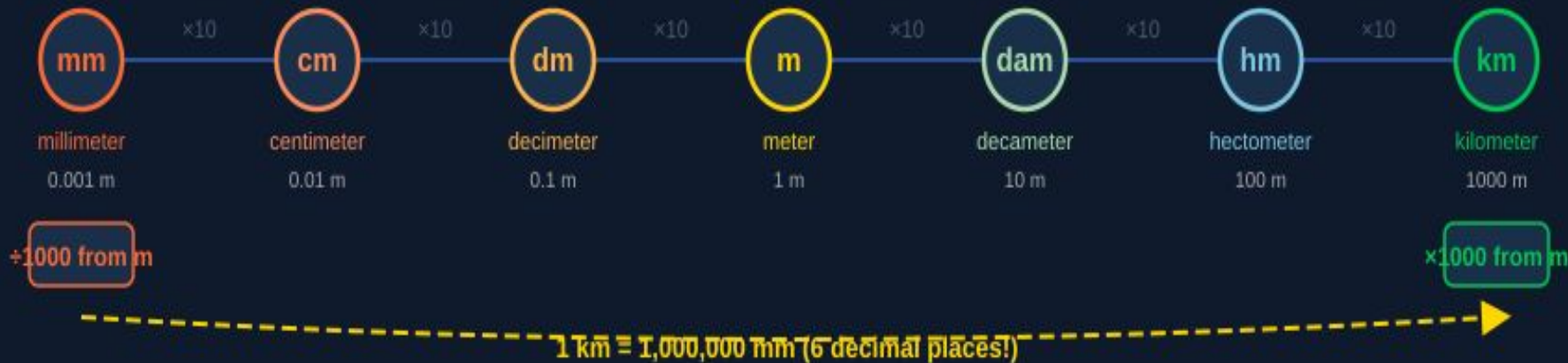
PREFIX	SYMBOL	MEANING	EXAMPLE	NOTE
kilo	k	$\times 1000$	kilometer (km)	YOU KNOW THIS
hecto	h	$\times 100$	hectometer (hm)	hectare (land area)
deca	da	$\times 10$	decameter (dam)	rarely used
(base)	m	$\times 1$	meter	the standard
deci	d	$\div 10$	decimeter (dm)	used in Europe
centi	c	$\div 100$	centimeter (cm)	YOU KNOW THIS
milli	m	$\div 1000$	millimeter (mm)	YOU KNOW THIS

Dimmer rows = rarely used in real life. But now you know something most adults don't!

The Complete Length Scale

Each step multiplies by 10. From mm to km is 6 steps — or 6 decimal places.

The Complete Length Scale — each step is $\times 10$



Quick quiz:

How many cm in 1 km?
1 dm = how many mm?
How many mm in 1 km?

Answers:

100,000 cm · 100 mm · 1,000,000 mm
(1 million millimeters!)

The Electronics Voltage Scale

Same structure as length — but in electronics, we skip the middle prefixes.

The Electronics Voltage Scale — same structure, different usage



Why do we skip cV, dV, daV, hV?

Electronics components operate in ranges that make those middle prefixes useless. A battery is 1.5V. An LED uses 20mA. A resistor might be 10k Ω . Nothing useful lands in between — so we skip straight from milli to base to kilo.

Moving the Decimal — Going RIGHT →

Converting from a base unit to milli: move decimal 3 places to the right

Moving Decimal RIGHT → number gets BIGGER

Use when converting: base unit → milli (e.g. Amps → milliamps)

Move decimal 3 places to the RIGHT → $\times 1000$

A → mA

0 . 0 1 9



3 places →

0.019 A = 19 mA

V → mV

0 . 5 0 0



3 places →

0.5 V = 500 mV

A → mA

0 . 0 4 7



3 places →

0.047 A = 47 mA

Going RIGHT makes the number BIGGER. 0.019 A → 19 mA. The value didn't change — just how we write it.

Moving the Decimal — Going LEFT



Converting from a base unit to kilo: move decimal 3 places to the left

Moving Decimal LEFT ← number gets SMALLER

Use when converting: base unit → kilo (e.g. ohms → kilohms)

Move decimal 3 places to the LEFT ← +1000

$\Omega \rightarrow \text{k}\Omega$

4	7	0	0	.
---	---	---	---	---

← 3 places

4700 Ω = 4.7 k Ω

$\text{mA} \rightarrow \text{A}$

1	0	0	0	.
---	---	---	---	---

← 3 places

1000 mA = 1 A

$\Omega \rightarrow \text{k}\Omega$

2	2	0	0	.
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← 3 places

2200 Ω = 2.2 k Ω

Going LEFT makes the number SMALLER. 4700 Ω → 4.7k Ω . Same resistor — just written in kilohms.

Chaining: 6 Decimal Places

What if you need to convert across two steps? You can chain them together.

Chaining Conversions — 6 Decimal Places

How many millimeters in 1 kilometer?

1 km

→
×1000

1000 m

→
×1000

1,000,000 mm

3 decimal places + 3 more decimal places = 6 decimal places total

Same idea — how many millivolts in 1 kilovolt?

1 kV

→
×1000

1000 V

→
×1000

1,000,000 mV

Same math — same answer. Prefixes work identically for every unit.

Fun fact: 1 km = 1,000,000 mm — one MILLION millimeters!

Voltage: Volts and Millivolts



AA battery

1.5 V

= 1500 mV

USB / Tinkercad

5 V

= 5000 mV

Red LED forward V

~2 V

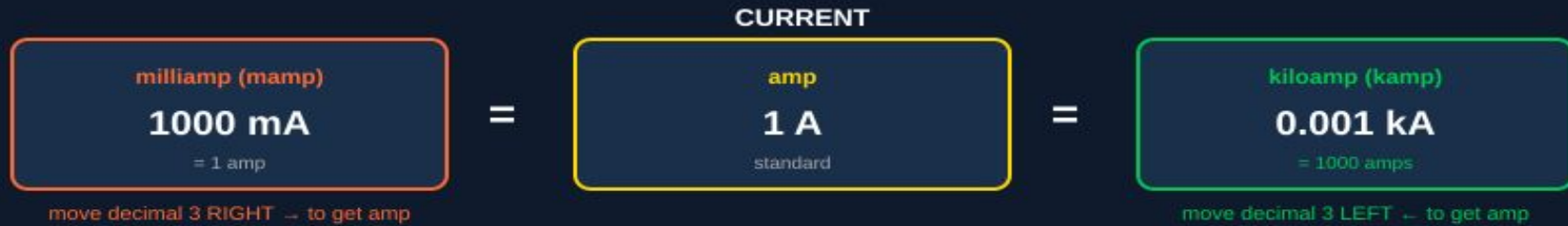
= 2000 mV

Microbit pin out

3.3 V

= 3300 mV

Current: Amps and Milliamps



⚡ In electronics, milliamps (mA) is more common than amps (A)

A typical LED uses ~20 mA = 0.020 A — tiny! A household light bulb uses ~1 A = 1000 mA

0.010 A = ?

→ 10 mA

0.020 A = ?

→ 20 mA

250 mA = ?

→ 0.250 A

0.001 A = ?

→ 1 mA

What Does Your Multimeter Actually Show?

Practice reading this — you'll see it every class



means?



Reading 0.019 A — step by step

Step 1: Identify the unit → Amps (A)

The dial and display both say A

Step 2: To convert A → mA, move decimal 3 RIGHT

0.019 → 0 1 9 . = 19

Step 3: Result

0.019 A = 19 mA

typical LED current!

Rule: ammeter shows Amps. To get milliamps, move decimal 3 RIGHT. 0.015 A = 15 mA · 0.023 A = 23 mA · 0.007 A = 7 mA

Resistance: Ohms and Kilohms

RESISTANCE — Ohms (Ω) and Kilohms ($k\Omega$)

To convert $\Omega \rightarrow k\Omega$: divide by 1000 (move decimal 3 LEFT)

Ohms (Ω)		Kilohms ($k\Omega$)	Typical use
220 Ω	→	0.22 $k\Omega$	LED resistor
470 Ω	→	0.47 $k\Omega$	LED resistor
1000 Ω	→	1 $k\Omega$	general use
4700 Ω	→	4.7 $k\Omega$	common value
10000 Ω	→	10 $k\Omega$	pullup / divider

Rule: 1000 Ω = 1 $k\Omega$ · move decimal 3 LEFT to convert $\Omega \rightarrow k\Omega$

Reading Resistor Values — What does "10k" mean?



220

220 Ω

two hundred
twenty ohms



470

470 Ω

four hundred
seventy ohms



1k

1,000 Ω

one kilohm



10k

10,000 Ω

ten kilohms



100k

100,000 Ω

one hundred
kilohms

k = kilo = $\times 1000$ → move decimal 3 LEFT

10k = $10 \times 1000 = 10,000 \Omega$ · 4.7k = $4.7 \times 1000 = 4,700 \Omega$ · 100k = 100,000 Ω

Worked Examples — Try These Together

Step through each one before the quiz

Worked Examples — Step by Step

0.020 A → mA

Which direction?

A → mA: move decimal 3 RIGHT

Calculate:

$$0.020 \times 1000 = 20 \text{ mA}$$

4700 Ω → kΩ

Which direction?

Ω → kΩ: move decimal 3 LEFT

Calculate:

$$4700 \div 1000 = 4.7 \text{ k}\Omega$$

3.3 V → mV

Which direction?

V → mV: move decimal 3 RIGHT

Calculate:

$$3.3 \times 1000 = 3300 \text{ mV}$$

0.047 A = ? mA

→ 47 mA

2200 Ω = ? kΩ

→ 2.2 kΩ

9 V = ? mV

→ 9000 mV

Quick Reference — Leave This Up During the Quiz

Unit	Full Name	milli- ($\div 1000$)	kilo- ($\times 1000$)
V	Volt	mV (millivolt)	kV (kilovolt — rare)
A	Amp	mA (milliamp) ← YOU'LL USE THIS	kA (rare)
Ω	Ohm	m Ω (rare)	kΩ (kilohm) ← YOU'LL USE THIS
m	Meter	mm (millimeter)	km (kilometer)

Memory trick: milli = mini (small, $\div 1000$) · kilo = king (big, $\times 1000$) · move decimal 3 places